EXPERIMENT 7

7	A college employs a large number of security personnel who are assigned duties at various locations across the campus. Each security person follows a duty schedule that includes date, duty start time, duty end time, and assigned location, ensuring that every campus location is covered. If a security person takes leave, it is recorded manually, and at the end of each month, salaries are calculated based on duties performed and leaves taken. Since manual handling of duty allocation, leave management, and salary calculation is time-consuming and complex, the college plans to develop an Online Security Management System to automate these activities. Develop Class Diagram, Use case Diagram and Activity Diagramto represent the workflow of the Online Security Management System.	Umbrello	CO1

Objective

To design UML diagrams for an Online Security Management System that automates duty allocation, leave management, and salary calculation of security personnel.

System Description (Short)

The system manages:

- Security personnel details
- Duty schedules
- Leave records
- Salary calculation

It automates manual processes for efficiency and accuracy.

🔹 PART 1 — CLASS DIAGRAM

Step 1: Open Umbrello

Open **Umbrello UML Modeller**
Click **File → New**

Step 2: Create Class Diagram

1. Right-click **Logical View**
2. Click **New Diagram → Class Diagram**
3. Rename as:

SecurityManagement_ClassDiagram

Step 3: Create Classes

Add these classes:

1. SecurityStaff
2. DutySchedule
3. Location
4. Leave
5. Salary
6. Admin

Step 4: Add Attributes and Methods

🔹 **SecurityStaff**

Attributes:

- staffId
- name
- phone
- designation Methods:

- viewDuty()
- applyLeave()

🔹 **DutySchedule**

Attributes:

- dutyId
- date
- startTime
- endTime

Methods:

- assignDuty()
- updateDuty()

🔹 **Location**

Attributes:

- locationId
- locationName Methods:
- updateLocation()

🔹 **Leave**

Attributes:

- leaveId
- leaveDate
- reason

Methods:

- approveLeave()

🔹 **Salary**

Attributes:

- salaryId
- month
- amount

Methods:

- calculateSalary()

🔹 **Admin**

Attributes:

- adminId
- name

Methods:

- assignDuty()
- manageLeave()
- generateSalary()

Step 5: Draw Relationship

Use Association / Aggregation:

From	To	Relationship
SecurityStaff	DutySchedule	Association
DutySchedule	Location	Association
SecurityStaff	Leave	Association
SecurityStaff	Salary	Association
Admin	SecurityStaff	Association

Step 6: Save Diagram

File → Save

🔹 PART 2 — USE CASE DIAGRAM

Step 1: Create Use Case Diagram

Right-click **Use Case View → New Diagram**
Rename as:
SecurityManagement_UseCaseDiagram

Step 2: Add Actors

Add:

- Admin
- Security Staff

Step 3: Add Use Cases

Admin Use Cases

- Assign Duty
- Update Duty
- Approve Leave
- Calculate Salary
- Generate Salary Report

Security Staff Use Cases

- View Duty Schedule
- Apply Leave
- View Salary

Step 4: Connect Actors to Use Cases

Use association lines between:

- Admin → admin use cases
- Security Staff → staff use cases

Step 5: Save Diagram

🔹 PART 3 — ACTIVITY DIAGRAM

Step 1: Create Activity Diagram

Right-click **Logical View → New Diagram → Activity Diagram**
Rename as:
SecurityManagement_ActivityDiagram

Step 2: Add Activities in Order

1. Start
2. Admin Assigns Duty
3. Security Staff Views Duty
4. Security Staff Performs Duty
5. Apply Leave (if required)
6. Admin Approves Leave
7. Monthly Duty Completion
8. Calculate Salary
9. Generate Salary
10. Security Staff Views Salary
11. Stop

Step 3: Add Decision Node

After **Apply Leave**

Decision:

Leave Approved?

- Yes → Continue
- No → Duty Continues

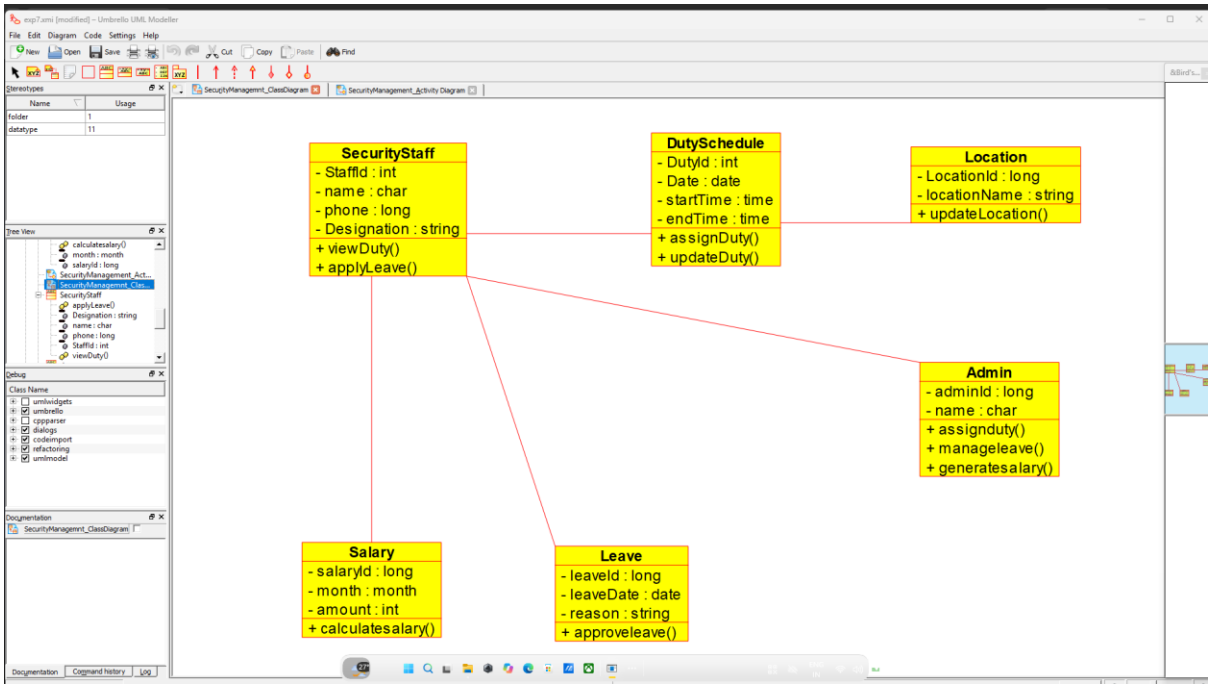
Step 4: Connect All Activities with Arrows

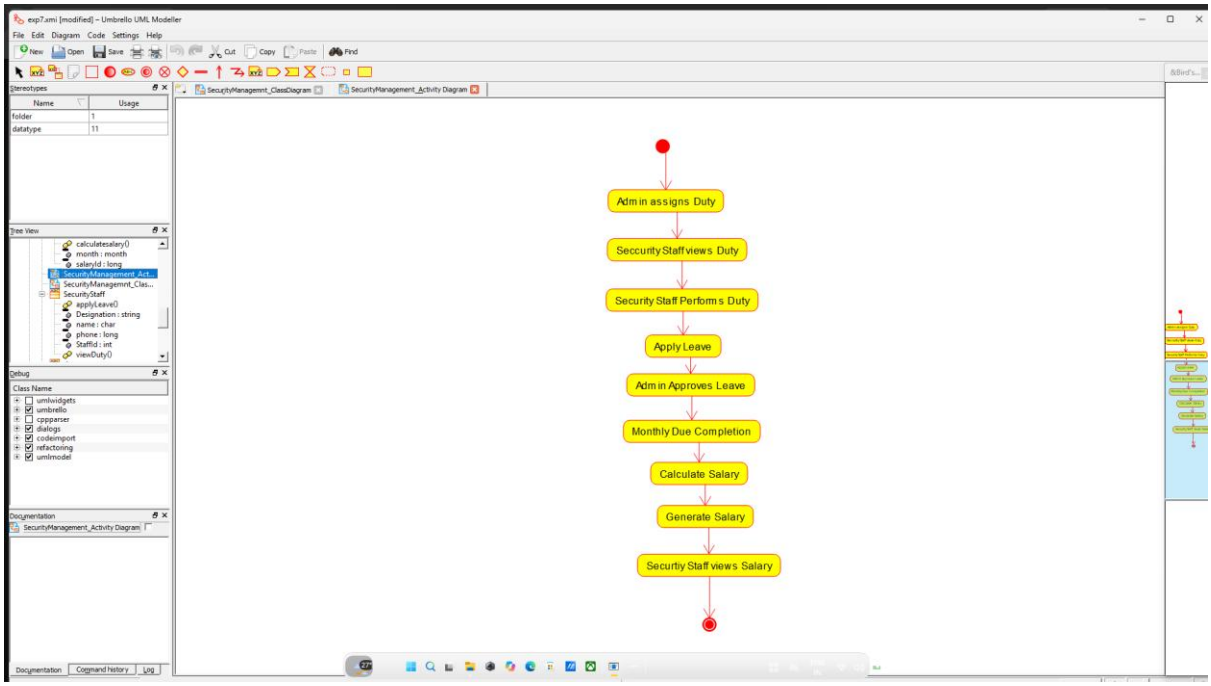
Use control flow arrows to show sequence.

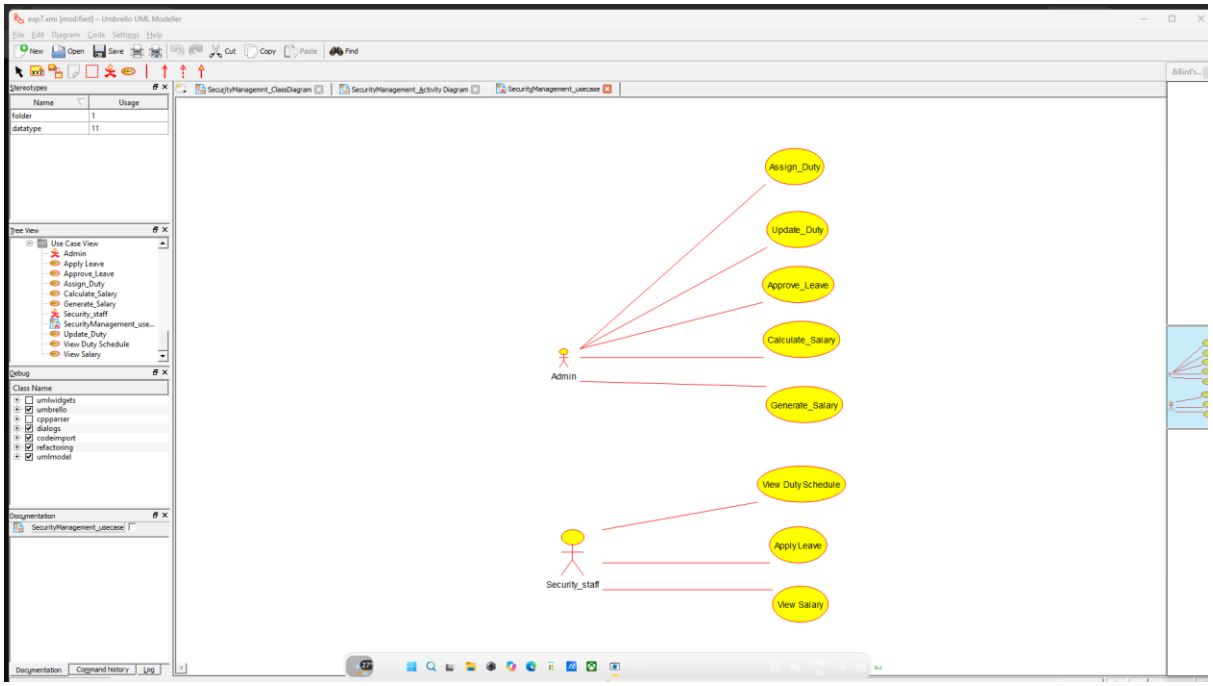
Step 5: Save Diagram

Result

Thus, Class Diagram, Use Case Diagram, and Activity Diagram for the Online Security Management System were successfully designed using Umbrello UML Modeller.


SecurityStaff
- StaffId : int
- name : char
- phone : long
- Designation : string
+ viewDuty()
+ applyLeave()
DutySchedule
- DutyId : int
- Date : date
- startTime : time
- endTime : time
+ assignDuty()
+ updateDuty()
Location
- LocationId : long
- locationName : string
+ updateLocation()
Admin
- adminId : long
- name : char
+ assignduty()
+ manageleave()
+ generatesalary()
Salary
- salaryId : long
- month : month
- amount : int
+ calculatesalary()
Leave
- leaveId : long
- leaveDate : date
- reason : string
+ approveleave()


Admin assigns Duty
Seccurity Staff views Duty
Security Staff Performs Duty
Apply Leave
Admin Approves Leave
Monthly Due Completion
Calculate Salary
Generate Salary
Securtiy Staff views Salary


Assign_Duty
Update_Duty
Approve_Leave
Calculate_Salary
Admin
Generate_Salary
View Duty Schedule
Apply Leave
Security_staff
View Salary